

Tune That TV

Surprisingly, TV Tuners haven't become as popular as they deserve to be. Here's the lowdown on what to buy and why

Jayesh Limaye

In the recently concluded India-Pakistan cricket series, cricket fever was at its pinnacle. You may think, "Wouldn't it be nice if I could capture and preserve those memorable moments (read: sixes, catches, wickets, or Afridi's superb annihilation of our bowling attack in the fifth ODI), so I could re-live the entire experience again?"

To add to your woes, that obsolete piece of hardware known as your VCR is no longer operational. Rejoice then, for it is that time of year again—time for the TV Tuner shootout.

TV tuners today have become so inexpensive and feature-packed that, by the time you are through reading this article, chances are you will have made up your mind to go and grab one that suits your needs.

This time around, we went looking for both internal and external TV tuners. External TV tuners were of two types—USB devices and set-top TV

boxes that are in-line gadgets that connect between the PC's analogue VGA-out and the VGA-in of your monitor.

Internal PCI TV Tuner Cards

This category is the most popular, mainly due to lower prices, good quality, features, and the fact that these cards sit inside the PC cabinet and don't occupy valuable space on your desktop.

We received 11 internal PCI TV Tuner cards. The companies that sent us cards included AVerMedia, Compro, Mercury, MSI, Pinnacle, PixelView and Tech-Com. MSI and Tech-Com were the newcomers this year.



PixelView's MediaCentre PlayTV@P3000

Features

We noticed a major shift in the manufacturers' choice of TV tuner chips this year. Last year, the Conexant (Brooktree) Bt8x8 chip dominated the TV tuner market. This year, the Philips 713x ranked high among most manufacturers.

All the cards came with bundled PVR software of some sort that supported at least MPEG-1 video capture. Time-shift was also supported by all the cards, with the exception of the Tech-Com Super TV SSD-TV-670.

Barring the Compro VideoMate TV Gold II, Pinnacle PCTV Stereo, MSI Mega TV Tuner card and the Tech-Com Super TV SSD-TV-670, all cards featured an FM radio tuner. The TV tuners based on the Philips SAA7130HL chip supported only mono TV audio. This really doesn't make a difference, since India does not have stereo cable TV signals.

The remote controllers that came with the TV tuners varied in size and functionality. Many came with remote controllers the size of a credit card. The Compro had a medium-sized remote controller, while the PixelView MediaCentre PlayTV@P3000's was big enough to club someone to death with, though it was definitely the most feature-rich of the lot—with exotic features such as zoom and DVD playback control.

How We Tested TV Tuners

We noted various features of the TV tuners, such as the chip, whether FM radio is supported, and whether the TV sound is stereo or mono. We noted the number of input/output ports, and higher points were awarded when a higher number of ports were featured.

We also rated the bundled accessories such as remote controller, cables and software packaging. Where PVR software was provided, its functionality, such as whether time-shifting was available, and whether MPEG-1/2/4 capture facility was provided was noted and rated accordingly. The ease of installation of the TV tuner was also noted and rated.

The test bench comprised an AMD Athlon 64 FX-53 processor on a MSI K8T Neo motherboard with 512 MB of Corsair DDR RAM running at 400 MHz and a 120 GB 7200 rpm Seagate Barracuda SATA/150 hard drive. The video card was a GeForce FX 5700LE, and the sound card was a Creative SoundBlaster Audigy 2 Platinum. We installed Windows XP SP1 and installed the necessary hotfixes. We then installed DirectX 9.0c and the latest drivers, and created a System Restore Point to get a reference point to which we could restore the system after testing each TV tuner.

This was done in order to avoid possible driver conflict issues, as many TV tuners that featured the same chip had drivers that consisted of similarly-named files and only different version numbers.

In the cable TV channel detection test, we deliberately created a cable TV point with a weak signal strength. This was done by introducing joints in the cable, which caused signal strengths to drop at each point. The TV viewing utility provided with the TV tuner was used to determine the number of channels that the TV tuner could detect.

After this, we rated the visual quality of the detected TV channels on a scale of 5. This was a subjective test, and we looked for artefacts and anomalies in different channels.

Next, we tested the TV tuners to capture content using both cable and S-Video as the input sources. We used VirtualDub 1.6.5 to capture content from the S-Video input in uncompressed raw AVI format (RGB-555) with PCM uncompressed audio.

We noted the number of frames dropped, and checked the quality of the captured content without any compression, made by either the hardware or software. We used VirtualDub, since it uses the minimum system resources.

We also used the TV tuners to capture regular cable content using the applications and software supplied by the manufacturers. This was done to check both, the quality of content, and also the real-world performance of the TV tuner. Any other issues, such as crashing of applications and drivers were also noted. In short, features, performance and pricing were considered to choose the winners.